

path in $G - \{u, v\}$. Note that Q too avoids B . Now $C \cup P \cup Q$ contains three cycles C_1, C_2, C_3 summing to C and each missing a vertex of C (Fig. 3.2.4). As every vertex of C sends an edge to B , we therefore have $b(C_i) > |B| = b(C)$ for every i , with the familiar contradiction. $\square$

3.3 Menger's theorem

The following theorem is one of the cornerstones of graph theory. It is another example of the particularly attractive type of theorem discussed in the notes for Chapter 2, a discussion which equally applies here:

Theorem 3.3.1. (Menger 1927)

Let $G = (V, E)$ be a graph and $A, B \subseteq V$. Then the minimum number of vertices separating A from B in G is equal to the maximum number of disjoint A – B paths in G .

[3.5.2]
[8.2.5]
[8.4.3]
[12.4.3]
[12.6.3]

We offer three proofs. Whenever G, A, B are given as in the theorem, we denote by $k(G, A, B)$ the minimum number of vertices separating A from B in G . Clearly, G cannot contain more than $k = k(G, A, B)$ disjoint A – B paths; our task will be to show that k such paths exist.

k

First proof. We apply induction on $\|G\|$. If G has no edge, then $|A \cap B| = k$ and we have k trivial A – B paths. So we assume that G has an edge $e = xy$. If G has no k disjoint A – B paths, then neither does G/e ; here, we count the contracted vertex v_e as an element of A (resp. B) in G/e if in G at least one of x, y lies in A (resp. B). By the induction hypothesis, G/e contains an A – B separator Y of fewer than k vertices. Among these must be the vertex v_e , since otherwise $Y \subseteq V$ would be an A – B separator in G . Then $X := (Y \setminus \{v_e\}) \cup \{x, y\}$ is an A – B separator in G of exactly k vertices.

We now consider the graph $G - e$. Since $x, y \in X$, every A – X separator in $G - e$ is also an A – B separator in G and hence contains at least k vertices. So by induction there are k disjoint A – X paths in $G - e$, and similarly there are k disjoint X – B paths in $G - e$. As X separates A from B , these two path systems do not meet outside X , and can thus be combined to k disjoint A – B paths. $\square$

Let $\mathcal{P}$ be a set of disjoint A – B paths, and let $\mathcal{Q}$ be another such set. We say that $\mathcal{Q}$ *exceeds* $\mathcal{P}$ if the set of vertices in A that lie on a path in $\mathcal{P}$ is a proper subset of the set of vertices in A that lie on a path in $\mathcal{Q}$, and likewise for B . Then, in particular, $|\mathcal{Q}| \geq |\mathcal{P}| + 1$.

exceeds

Second proof. We prove the following stronger statement:

If $\mathcal{P}$ is any set of fewer than $k = k(G, A, B)$ disjoint A – B paths in G , then G contains a set of $|\mathcal{P}| + 1$ disjoint A – B paths exceeding $\mathcal{P}$.

Keeping G and A fixed, we let B vary and apply induction on $|\bigcup \mathcal{P}|$. Let R be an A – B path that avoids the (fewer than k) vertices of B that lie on a path in $\mathcal{P}$. If R avoids all the paths in $\mathcal{P}$, then $\mathcal{P} \cup \{R\}$ exceeds $\mathcal{P}$, as desired. (This will happen when $\mathcal{P} = \emptyset$, so the induction starts.) If not, let x be the last vertex of R that lies on some $P \in \mathcal{P}$ (Fig. 3.3.1).

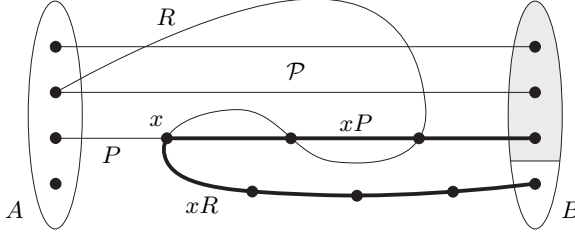


Fig. 3.3.1. Paths in the second proof of Menger's theorem

Put

$$B' := B \cup V(xP \cup xR) \quad \text{and} \quad \mathcal{P}' := (\mathcal{P} \setminus \{P\}) \cup \{Px\}.$$

Then $|\mathcal{P}'| = |\mathcal{P}|$ but $|\bigcup \mathcal{P}'| < |\bigcup \mathcal{P}|$, and $k(G, A, B') \geq k(G, A, B)$, so by the induction hypothesis there is a set $\mathcal{Q}'$ of $|\mathcal{P}'| + 1$ disjoint A – B' paths exceeding $\mathcal{P}'$. Then $\mathcal{Q}'$ contains a path Q ending at x , and a unique path Q' whose last vertex y is not among the last vertices of the paths in $\mathcal{P}'$.

If $y \notin xP$, we let $\mathcal{Q}$ be obtained from $\mathcal{Q}'$ by appending xP to Q , and appending yR to Q' if $y \notin B$. Otherwise $y \in xP$, and we let $\mathcal{Q}$ be obtained from $\mathcal{Q}'$ by appending xR to Q and yP to Q' . In all cases $\mathcal{Q}$ exceeds $\mathcal{P}$, as desired. $\square$

Applied to a bipartite graph, Menger's theorem specializes to the assertion of König's theorem (2.1.1). For our third proof, we shall adapt the alternating path proof of König's theorem to the more general set-up of Theorem 3.3.1. Let again G, A, B be given, and let $\mathcal{P}$ be a set of disjoint A – B paths in G . Let us say that an A – B separator $X \subseteq V$ lies on $\mathcal{P}$ if it consists of a choice of exactly one vertex from each path in $\mathcal{P}$. If we can find such a separator X , then clearly $k \leq |X| = |\mathcal{P}|$, and Menger's theorem will be proved.

Put

$$V[\mathcal{P}] := \bigcup \{V(P) \mid P \in \mathcal{P}\}$$

$$E[\mathcal{P}] := \bigcup \{E(P) \mid P \in \mathcal{P}\}.$$

$\mathcal{P}$

on

Let a walk $W = x_0 e_0 x_1 e_1 \dots e_{n-1} x_n$ in G with $e_i \neq e_j$ for $i \neq j$ be said to *alternate* with respect to $\mathcal{P}$ (Fig. 3.3.2) if it starts in $A \setminus V[\mathcal{P}]$ and the following three conditions hold for all $i < n$ (with $e_{-1} := e_0$ in (iii)):

W, x_i, e_i
alternating
walk

- (i) if $e_i = e \in E[\mathcal{P}]$, then W traverses the edge e backwards, i.e. $x_{i+1} \in P\overset{\circ}{x}_i$ for some $P \in \mathcal{P}$;
- (ii) if $x_i = x_j$ with $i \neq j$, then $x_i \in V[\mathcal{P}]$;
- (iii) if $x_i \in V[\mathcal{P}]$, then $\{e_{i-1}, e_i\} \cap E[\mathcal{P}] \neq \emptyset$.

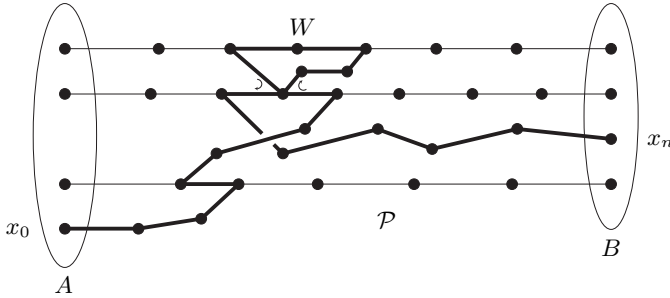


Fig. 3.3.2. An alternating walk from A to B

Note that, by (ii), any vertex outside $V[\mathcal{P}]$ occurs at most once on W . And since the edges e_i of W are all distinct, (iii) implies that any vertex $v \in V[\mathcal{P}]$ occurs at most twice on W . For $v \neq x_n$, this can happen in exactly the following two ways. If $x_i = x_j$ with $0 < i < j < n$, then

$$\begin{aligned} &\text{either } e_{i-1}, e_j \in E[\mathcal{P}] \text{ and } e_i, e_{j-1} \notin E[\mathcal{P}] \\ &\text{or } e_i, e_{j-1} \in E[\mathcal{P}] \text{ and } e_{i-1}, e_j \notin E[\mathcal{P}]. \end{aligned}$$

Unless otherwise stated, any use of the word ‘alternate’ below will refer to our fixed path system $\mathcal{P}$.

The next two lemmas together make up our third proof of Menger’s theorem. We state and prove them in a way that makes them reusable in Chapter 8, when we prove Menger’s theorem for infinite graphs.

Lemma 3.3.2. *If an alternating walk W as above ends in $B \setminus V[\mathcal{P}]$, then G contains a set of disjoint A – B paths exceeding $\mathcal{P}$.*

[8.4.7]

Proof. We may assume that W has only its first vertex in $A \setminus V[\mathcal{P}]$ and only its last vertex in $B \setminus V[\mathcal{P}]$. Let H be the graph on $V(G)$ whose edge set is the symmetric difference of $E[\mathcal{P}]$ with $\{e_0, \dots, e_{n-1}\}$. In H , the ends of the paths in $\mathcal{P}$ and of W have degree 1 (or 0, if the path or W is trivial), and all other vertices have degree 0 or 2.

For each vertex $a \in (A \cap V[\mathcal{P}]) \cup \{x_0\}$, therefore, the component of H containing a is a path, $P = v_0 \dots v_k$ say, which starts in a and ends in A or B . Using conditions (i) and (iii), one easily shows by induction

on $i = 0, \dots, k-1$ that P traverses each of its edges $e = v_i v_{i+1}$ in the forward direction with respect to $\mathcal{P}$ or W . (Formally: if $e \in P'$ with $P' \in \mathcal{P}$, then $v_i \in P' \hat{v}_{i+1}$; if $e = e_j \in W$, then $v_i = x_j$ and $v_{i+1} = x_{j+1}$.) Hence, P is an A – B path. (When G is infinite, this last conclusion uses the fact that W meets only finitely many paths in $\mathcal{P}$, and hence every component of H is finite.)

Similarly, for every $b \in (B \cap V[\mathcal{P}]) \cup \{x_n\}$ there is an A – B path in H that ends in b . The set of A – B paths in H therefore exceeds $\mathcal{P}$. $\square$

[8.4.7] **Lemma 3.3.3.** *If no alternating walk W as above ends in $B \setminus V[\mathcal{P}]$, then G contains an A – B separator on $\mathcal{P}$.*

Proof. Let

$$A_1, A_2 \quad A_1 := A \cap V[\mathcal{P}] \quad \text{and} \quad A_2 := A \setminus A_1,$$

and

$$B_1, B_2 \quad B_1 := B \cap V[\mathcal{P}] \quad \text{and} \quad B_2 := B \setminus B_1.$$

x_P For every path $P \in \mathcal{P}$, let x_P be the last vertex of P that lies on some alternating walk; if no such vertex exists, let x_P be the first vertex of P . Our aim is to show that

$$X \quad X := \{x_P \mid P \in \mathcal{P}\}$$

meets every A – B path in G ; then X is an A – B separator on $\mathcal{P}$.

Q Suppose there is an A – B path Q that avoids X . We know that Q meets $V[\mathcal{P}]$, as otherwise it would be an alternating walk ending in B_2 . Now the A – $V[\mathcal{P}]$ path in Q is either an alternating walk or consists only of the first vertex of some path in $\mathcal{P}$. Therefore Q also meets the vertex set $V[\mathcal{P}']$ of

$$\mathcal{P}' \quad \mathcal{P}' := \{Px_P \mid P \in \mathcal{P}\}.$$

y, P, x Let y be the last vertex of Q in $V[\mathcal{P}']$, say $y \in P \in \mathcal{P}$, and let $x := x_P$. As Q avoids X and hence x , we have $y \in P \hat{x}$. In particular, $x = x_P$ is not the first vertex of P , and so there is an alternating walk W ending at x . Then $W \cup xPyQ$ is a walk from A_2 to B (Fig. 3.3.3). If this walk alternates and ends in B_2 , we have our desired contradiction.

W

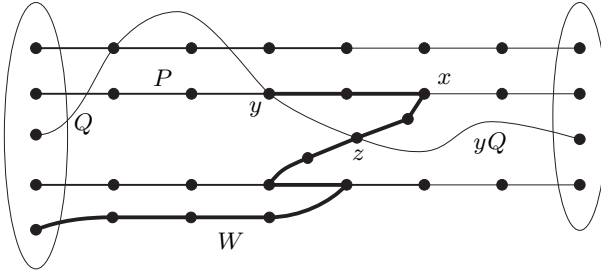


Fig. 3.3.3. Alternating walks in the proof of Lemma 3.3.3.

How could $W \cup xPyQ$ fail to alternate? For example, W might already use an edge of xPy . But if x' is the first vertex of W on xPy , then $W' := Wx'Py$ is an alternating walk from A_2 to y . (By Wx' we mean the initial segment of W ending at the first occurrence of x' on W ; from there, W' follows P back to y .) Even our new walk $W'yQ$ need not yet alternate: for example, W' might still meet yQ . By definition of $\mathcal{P}'$ and W , however, and the choice of y on Q , we have

$$V(W') \cap V[\mathcal{P}] \subseteq V[\mathcal{P}'] \quad \text{and} \quad V(yQ) \cap V[\mathcal{P}'] = \emptyset.$$

Thus, W' and yQ can meet only outside $\mathcal{P}$.

If W' does indeed meet yQ , we let z be the first vertex of W' on yQ and set $W'' := W'zQ$. Otherwise we set $W'' := W' \cup yQ$. In both cases W'' alternates with respect to $\mathcal{P}'$, because W' does and yQ avoids $V[\mathcal{P}']$. (W'' satisfies condition (iii) at y in the second case even if y occurs twice on W' , because W'' then contains the entire walk W' and not just its initial segment $W'y$.) By definition of $\mathcal{P}'$, therefore, W'' avoids $V[\mathcal{P}] \setminus V[\mathcal{P}']$. Thus W'' also alternates with respect to $\mathcal{P}$ and ends in B_2 , contrary to our assumptions. $\square$

Third proof of Menger's theorem. Let $\mathcal{P}$ contain as many disjoint A – B paths in G as possible. Then by Lemma 3.3.2, no alternating walk ends in $B \setminus V[\mathcal{P}]$. By Lemma 3.3.3, this implies that G has an A – B separator X on $\mathcal{P}$, giving $k \leq |X| = |\mathcal{P}|$ as desired. $\square$

A set of a – B paths is called an a – B *fan* if any two of the paths have only a in common.

Corollary 3.3.4. *For $B \subseteq V$ and $a \in V \setminus B$, the minimum number of vertices separating a from B in G is equal to the maximum number of paths forming an a – B fan in G .*

Proof. Apply Theorem 3.3.1 to $G - a$ with $A := N_G(a)$. $\square$

Corollary 3.3.5. *Let a and b be two distinct vertices of G .*

- (i) *If $ab \notin E$, then the minimum number of vertices separating a from b in G is equal to the maximum number of independent a – b paths in G .*
- (ii) *The minimum number of edges separating a from b in G is equal to the maximum number of edge-disjoint a – b paths in G .*

Proof. (i) Apply Theorem 3.3.1 to $G - \{a, b\}$, with $A := N_G(a)$ and $B := N_G(b)$.

(ii) Apply Theorem 3.3.1 to the line graph of G , with $A := E(a)$ and $B := E(b)$. $\square$

[4.2.7]
[6.6.1]
[9.4.2]

Theorem 3.3.6. (Global Version of Menger's Theorem)

- (i) A graph is k -connected if and only if it contains k independent paths between any two vertices.
- (ii) A graph is k -edge-connected if and only if it contains k edge-disjoint paths between any two vertices.

Proof. (i) If a graph G contains k independent paths between any two vertices, then $|G| > k$ and G cannot be separated by fewer than k vertices; thus, G is k -connected.

Conversely, suppose that G is k -connected (and, in particular, has more than k vertices) but contains vertices a, b not linked by k independent paths. By Corollary 3.3.5 (i), a and b are adjacent; let $G' := G - ab$. Then G' contains at most $k - 2$ independent a - b paths. By Corollary 3.3.5 (i), we can separate a and b in G' by a set X of at most $k - 2$ vertices. As $|G| > k$, there is at least one further vertex $v \notin X \cup \{a, b\}$ in G . Now X separates v in G' from either a or b – say, from a . But then $X \cup \{b\}$ is a set of at most $k - 1$ vertices separating v from a in G , contradicting the k -connectedness of G .

(ii) follows straight from Corollary 3.3.5 (ii). $\square$

a, b
 G'
 X
 v

3.4 A -paths and Mader's theorem

In analogy to Menger's theorem we may consider the following questions. Given a set A of vertices in a graph, how many disjoint, edge-disjoint, or independent A -paths can we find in that graph? Is the largest number of such paths related to the least size of a vertex or edge cover of all its A -paths?

An early theorem of Gallai goes in this direction. In Erdős-Pósa spirit – see Chapter 2.3 – it says that if a graph fails to contain many disjoint A -paths it has a small set of vertices that covers all its A -paths, and thus 'totally disconnects' A :

Theorem 3.4.1. (Gallai 1961)

Let A be a set of vertices in a graph G , and $k \geq 1$ an integer. Then G either contains k disjoint A -paths or has a set of at most $2k - 2$ vertices that meets every A -path.

The graph K^{2k-1} , with A its entire vertex set, shows that $2k - 2$ is lowest possible as a general bound, one that depends only on k .

Theorem 3.4.1, as well as its analogue for edge-disjoint paths (with the same bound of $2k - 2$), will be an easy consequence of Mader's theorem below, via its Corollary 3.4.3. See Exercises 25 and 30.